

Homework (Habanero)

- Q1.** Factor the expression: $x^2 + 5x + 4x + 20$
(A) $x(x + 5) + 4(x + 5)$
(B) $x(x + 4) + 5(x + 4)$
(C) $x^2 + 9x + 20$
(D) $x(x + 4) + 5x + 4$
(E) $x(x + 3) + 4(x + 5)$
- Q2.** A nurse administers a dosage of $2x + 6$ mg of medicine to a patient every $x + 3$ hours. Factor the expression: $(2x + 6) + (x + 3)(2)$
(A) $(2x + 6) + 2(x + 3)$
(B) $(x + 3)(2x + 2)$
(C) $(2x + 6)(x + 3)$
(D) $2(x + 3)^2$
(E) $(x + 3)(2x + 8)$
- Q3.** Factor the expression: $x^2 + 7x + 2x + 14$
(A) $x(x + 7) + 2(x + 7)$
(B) $x(x + 2) + 7(x + 2)$
(C) $x^2 + 9x + 14$
(D) $x(x + 2) + 7x + 2$
(E) $x(x + 3) + 4(x + 7)$
- Q4.** A doctor prescribes a treatment that increases cell growth by $3x + 12$ cells per day, and another treatment that increases cell growth by $x + 4$ cells every 3 days. Factor the expression: $(3x + 12) + 3(x + 4)$
(A) $3(3x + 4) + 12$
(B) $(x + 4)(3x + 3)$
(C) $3(x + 4)^2$
(D) $(3x + 12)(x + 4)$
(E) $3(x + 4) + (3x + 12)$
- Q5.** Factor the expression: $x^2 + 9x + 4x + 36$
(A) $x(x + 9) + 4(x + 9)$
(B) $x(x + 4) + 9(x + 4)$
(C) $x^2 + 13x + 36$
(D) $x(x + 4) + 9x + 4$
(E) $x(x + 3) + 4(x + 9)$
- Q6.** A medical researcher collects data on the number of patients who respond to a new treatment, and finds that $2x + 10$ patients respond within $x + 5$ days. Factor the expression: $(2x + 10) + (x + 5)(2)$
(A) $(2x + 10) + 2(x + 5)$
(B) $(x + 5)(2x + 2)$
(C) $(2x + 10)(x + 5)$
(D) $2(x + 5)^2$
(E) $(x + 5)(2x + 10)$
- Q7.** Factor the expression: $x^2 + 6x + 3x + 18$
(A) $x(x + 6) + 3(x + 6)$
(B) $x(x + 3) + 6(x + 3)$
(C) $x^2 + 9x + 18$
(D) $x(x + 3) + 6x + 3$
(E) $x(x + 3) + 6(x + 3)$
- Q8.** A pharmacist dispenses a medication that has a $4x + 16$ mg dosage, which needs to be taken every $x + 4$ hours. Factor the expression: $(4x + 16) + (x + 4)(4)$
(A) $(4x + 16) + 4(x + 4)$
(B) $(x + 4)(4x + 4)$
(C) $(4x + 16)(x + 4)$
(D) $4(x + 4)^2$
(E) $(x + 4)(4x + 16)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression: $x^2 + 8x + 6x + 48$. Verify your answer by multiplication.
- Q10.** A biologist studies the growth rate of bacteria, which can be modeled by the expression: $(3x + 12) + (x + 4)(3)$. Factor this expression and interpret the results in the context of the problem.

Chef's Tips

- Always verify factoring by multiplication
- Use the distributive property to check your work
- Pay attention to the context of word problems to ensure accurate interpretation